

CHAPTER-1

HISTORY OF OIL

Oil is known to mankind since ages. In fact man has been obtaining crude oil and using it for various purposes like medical, fuel, plastering, checking leakages, warfare, etc. Chinese were first to use oil and gave the world the first drilling machine, but they were not drilling for oil. In the land locked areas of China, they used to drill for brine, to make salt .. Sometimes they would encounter crude oil which was an unwanted material, however they knew the uses of oil and they used it as medicine. Bamboo drilling ,was known in China as early as 1637. The Romans used crude oil in warfare, the Egyptians used for preservation of Mummies and Indians used it in Ayurved as Shilajit for medical purpose.

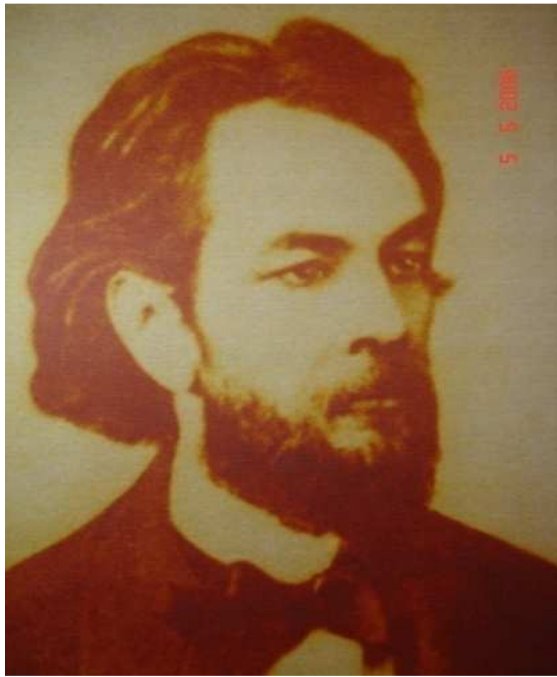


Drilling of deep wells in ancient China

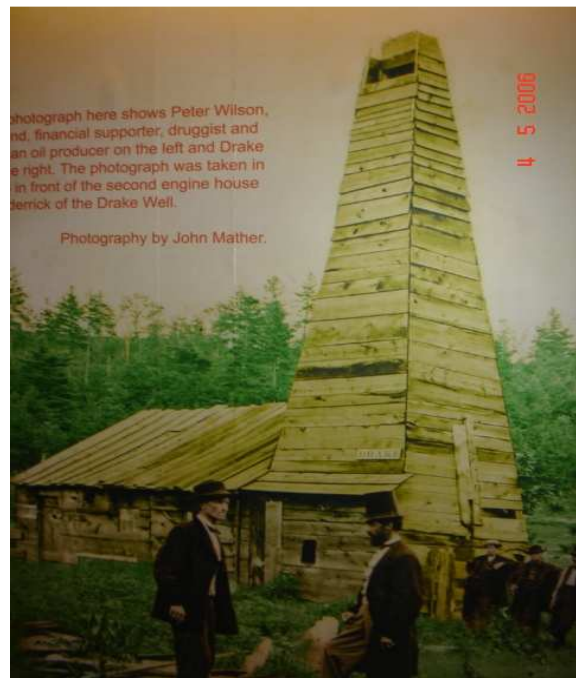


Use of crude oil for medicinal purpose in ancient times

The history of modern oil starts with George Bissell who is also known as the father of modern oil Industry. Around 1850 whale oil was used for lighting purposes, in the west. Due to indiscriminate hunting, the whales were becoming scarce and whale oil was becoming costlier, and people were looking for alternatives. George Bissell thought that the rock oil could be used for this purpose, so he engaged some scientists to analyse and find its applications. The scientists advised him that it was a very useful substance and there were a number of products that could be obtained out of it, including a liquid for lighting. He got encouraged and decided to deliberately drill for oil and produce it, not simply obtain it from seepages. So he engaged a Railroad conductor named Edwin Laurentine Drake and sent him over to Titusville in Pennsylvania, USA to drill for oil. But those days going anywhere and drilling was not easy, as the locals were hostile. So George Bissell called him Colonel Drake to impress the people who may get scared of the military title attached to his name.



George Bissell

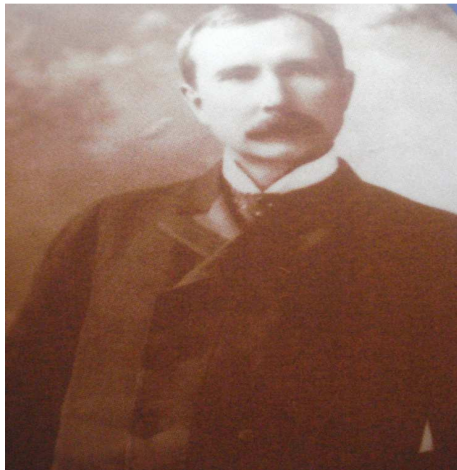


Drake oil well

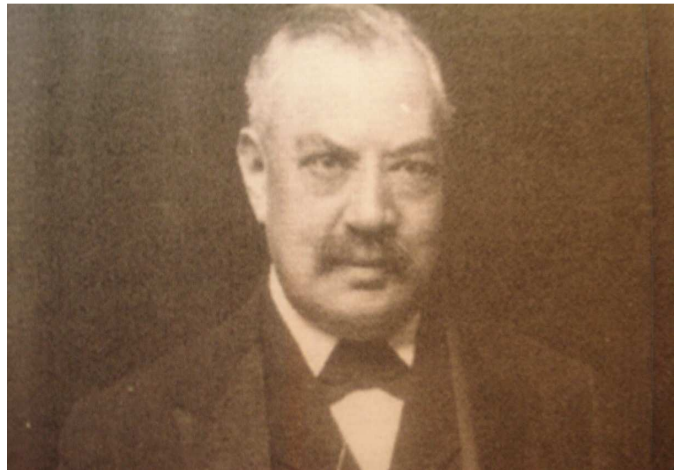
When Drake went and started this drilling operation, he met with the lot of difficulties. The money was running out and he was asked for more money to be sent by George Bissell. At that point of time on a weekend Drake and his men got frustrated and went away leaving the drilling equipment at the site of work. When they came back after the weekend they found oil filled in the pit. This was the first oil well drilled in 1859 where the oil was struck at a depth of 69 feet only.

But unfortunately though Drake invented the drilling rig, he never patented it. So towards later years of his life he lived on the pension from the government, a life of penury.

As the demand for oil started increasing there were several players who came in this field. John D. Rockefeller was a towering personality amongst all of them who established the Standard Oil Company in 1870. It was not only for the oil business, but the way he did the business, and the way he monopolized and controlled the entire oil business of the world, that he has become famous. Abraham Gesner invented the kerosene lamp in 1850 which began to be run on the processed crude oil instead of whale oil and was used to provide light during the long winter nights in Europe and America. For quite sometime the processed crude oil became the main source of lighting and there was big business in oil. However, there was a very brief period of recession due to the invention of electric bulb by Thomas Elva Edison. But, in the meantime due to the discovery of internal combustion engine once again oil became an important commodity and was now used for transportation purpose. There were many other players who became a part of this business.. The Noble Brothers started the business from Baku region in Europe. John Gaining started it in Texas. Marcus Samuel an intelligent businessman from Britain challenged the Standard Oil Company by starting the Shell Transport and Trading Company, which was mainly involved in trading in oil and extended his business to far fetched regions of the world such as the Asian market. Shell is the largest oil company in the world, today. As the importance of oil started increasing, the oil business started becoming more and more politicized.



1870: John D. Rockefeller started the Standard Oil company- the first real multinational.

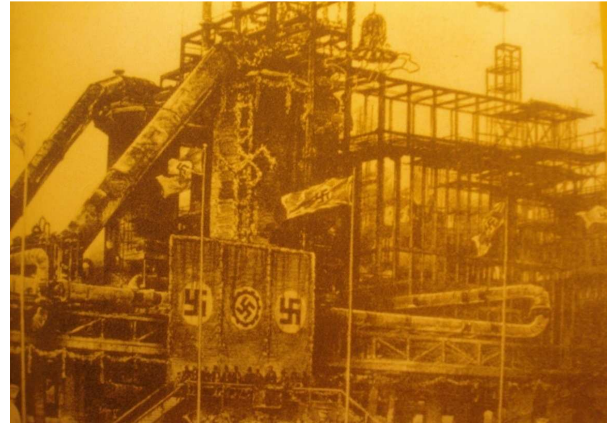


Marcus Samuel in 1892 broke the monopoly of Standard Oil in the world Oil market. He established the "Shell" Transport and Trading Company Limited

The Oil Companies were like parallel governments. They started making agreements between themselves, sharing the oil fields in the different parts of the world. The entire Arab region was divided by the British oil companies through an agreement called the Red Line Agreement which decided ,that who will produce, from where, how much quantities, ,and how it would be marketed. Oil started becoming more and more important for political decisions.



Adolf Hitler made oil central to his plan for Conquest in World war II.



The synthetic fuel plant -1937. Such fuels provided more than half of Germany's total oil supply during the war

Adolf Hitler , Eisenhower and many world leaders talked about the importance of oil in their strategy. Unfortunately in this entire game, a few companies and a few countries that were having the technology and could hold on the oil business were benefiting. The countries where the oil was actually available and was being produced were not getting their due share. So the Saudi Arabia and Venezuela oil ministers along with the other oil producing nations of the Middle East such as Iran ,Iraq and Kuwait got together and formed OPEC (Organization of Petroleum Exporting Countries) in 1960. Today it is the OPEC which determines the oil prices and has the maximum influence on the global politics. Oil has gained such an importance that in the past one century most wars in the world have been fought

directly or indirectly for crude oil. Germany, Japan and Italy, the three Axis powers did not possess oil fields which was one of the reasons for the World War II.



The great "Oil hunt" after the World war I led to the Oil boom in Venezuela



Yom Kippur war sent gasoline prices soaring and Americans into gas lines. The same happened in 1979 when Shah of Iran fell from power.

HISTORY OF OIL INDUSTRY IN INDIA

India has not been much behind the world in the search for oil The first oil well was drilled in Makum in Assam in 1866, but unfortunately it did not bear any oil. In 1889 Assam Railway and Trading Company started drilling a well at Digboi and it struck oil at a depth of 172 feet. In 1902 a refinery was set up at Digboi for processing this crude. Today this refinery has been modernized and owned by the Indian Oil Corporation



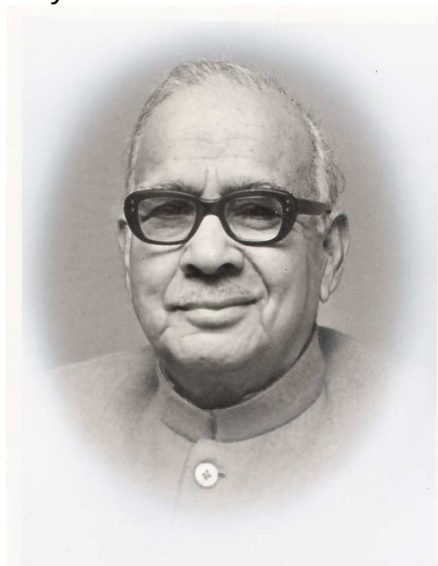
Oil was struck at Digboi Assam in 1889



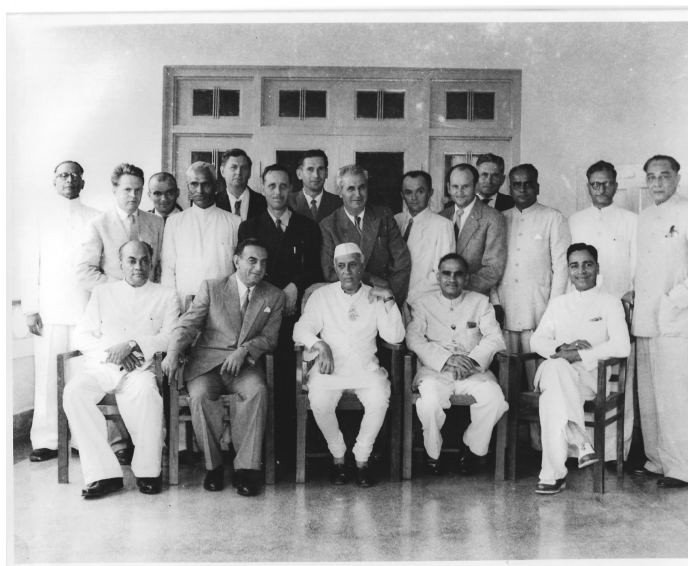
Refinery was set up at Digboi in 1902

.After independence the importance of oil for economic independence was realized by the then Prime minister Pt. Jawaharlal Nehru. He encouraged Shri K.D. Malviya to take up the challenge of oil exploration in the country. K.D. Malviya a visionary and having a good knowledge of the oil industry, was responsible for starting Oil and Natural Gas Commission in India on 14th August 1956, with himself as its first Chairman. Russians were the main source of technology, support and help for ONGC in the formative years because most of the western countries were of the opinion there was no oil in India. Despite all the resistance Malaviya faced from different quarters, he continued working with a conviction to achieve his objectives, with the support of Jawaharlal Nehru. Soon he proved his critics to be wrong, with the success of striking oil at Cambay in 1958, Rudrasagar in Assam in 1960 and a

major oil field at Ankleswar in 1960. Oil was found at the offshore field at Bombay High in 1974 and was transported by an oil tanker called Jawaharlal Nehru on 21st May 1976.



Visionary first Chairman of ONGC -Shri Keshava Dev Malaviya-



Pt. Jawahar Lal Nehru, with the oil men of the country at the Patiala House, Dehradun.

Today ONGC has emerged as the most valuable company of the country and is truly multi-national which has its properties in different parts of the world.

IMPORTANT EVENTS/LANDMARKS IN THE WORLD OIL HISTORY :

- ❖ Olive oil, Beeswax, Whale oil, Sesame oil, Nut oil - fuel till late 18th century.
- ❖ Ancient cultures - crude oil from seepages as sealant; 5000 years ago Sumerians, asphalt as mosaics, Mesopotamians - bitumen - line water canals, check leakage in boats and build roads.
- ❖ Chinese - first to discover oil deposits in salt wells.
- ❖ Romans used Flaming containers of oil as weapons of war.
- ❖ In 13th century, Marco Polo saw oil being collected from seepages.
- ❖ 16th century – Baku - hand-dug holes
- ❖ Egyptians used crude oil to preserve mummies
- ❖ Cure for Asthma, Skin Diseases and Arthritis- Chinese and Red Indians
- ❖ Shilajeet used in Ayurved
- ❖ Abraham Gesner Invented kerosene Lamp in 1850 - Kerosene synthesized from Coal
- ❖ **1855: George Bissel, James Townsend engaged Yale University chemist Benjamin Silliman Jr., to analyse crude oil and subsequently formed the Pennsylvania Rock oil Company**
- ❖ 1859: First oil well 691/2 feet at Titusville Pennsylvania - drilled by Col. Laurentine Drake
- ❖ 1870: John D. Rockefeller started Standard Oil company- the first real multinational.
- ❖ The Nobel brothers and Rothschilds were in oil business in the Baku oil
- ❖ Marcus Samuel in 1892 broke the monopoly of Standard Oil in the world Oil market. He established the “Shell” Transport and Trading Company Limited
- ❖ 1876: Nicholas August Otto - four-stroke IC engine.
- ❖ 1878: Thomas Alva Edison - electric light bulb. Oil industry – brief recession
- ❖ 1886: Karl Benz & Wilhelm Diamler - first automobile powered by IC engine

- ❖ 1903 : Wright Brothers motored and manned airplane
- ❖ 1908 : Henry Ford sells first Model T car.
- ❖ The oil discovery at the jungles of Telega on the island of Sumatra made the Royal Dutch to enter the world oil business in 1890
- ❖ 1911 : Winston Churchill decided to change Royal Navy's Feedstock from Coal to Oil.
- ❖ 1913 - World producing only 50 million tons of oil.
- ❖ 1914-1919: World war-1, tanks invented, aerial fighting.
- ❖ War research gave new technologies to the oil industry.
- ❖ In September 1914 as the German army closed in on Paris, city's taxi-cabs were used to transport troops to the front. Oil had gone to war
- ❖ 1896 : Hungarian Scholar - Eotvos torsion balance measure gravity anomalies – birth of Geophysics.
- ❖ 1912 : Conrad Schlumberger invents wireline resistivity logging.
- ❖ 1922 Mintrop - German Engineer - process for locating enemy artillery - birth of Seismic - a cap rock in North Germany discovered by seismic refraction.
- ❖ In 1908 oil was discovered in Persia.
- ❖ In 1920 oil discovery in Oklahoma-USA
- ❖ In 1930 oil struck at Texas USA
- ❖ Venezuelan oil boom-1922
- ❖ 1968- Alaska oil finding
- ❖ By 1920s, "Seven Sisters"- Esso, Mobil, Chevron, Royal Dutch, Shell, Texaco, Gulf and British Petroleum - stranglehold on the oil business - beginning of the cartelisation of oil business. Oil companies - got together at Achnacherry Island in England to act in "unison" in all matters regarding oil.
- ❖ 1922-28-The Red line Agreement
- ❖ On 26th July 1956 Nasser nationalized the Suez Canal. France and Egypt jointly financed and opened the canal in 1869. Later the share of Egypt was bought by the British. Nationalization of canal led to a combined attack by Israel, France and Britain resulting in world oil crisis. Nasser was the winner – Canal Came under the total control of Egypt
- ❖ In 1959 Eisenhower administration imposed ration on foreign oil imports to United States which hit Venezuela and other middle-east Oil Exporting Countries leading to the formation of OPEC
- ❖ OPEC – founded in 1960
- ❖ The six days war- 1967 Arab-Israel war was fought in 1967 between Israel and Arab neighbours Egypt, Jordan and Syria. As a result of the war Suez canal was closed leading to world oil crisis
- ❖ On October 6, 1973 Yom Kippur, the holiest day in the Jewish calendar, Egypt and Syria opened a coordinated surprise attack against Israel. Later in the war Iraq, Saudi Arabia and Kuwait also joined. During this period the Arab oil ministers agreed for a secret oil embargo, thus cutting oil production by 5% every month to the last month level, which led to a direct oil crisis of western countries, especially USA. This period saw the solidarity of the OPEC countries – the first oil shock. Yom Kippur war sent gasoline prices soaring and Americans into gas lines. The same happened in 1979 when Shah of Iran fell from power.
- ❖ The Gulf war- 1990-91 Saddam Hussein, the Iraqi strongman who triggered the first post-Cold war oil crisis. Iraqi army set 600 oil wells on fire

IMPORTANT EVENTS/LANDMARKS IN THE INDIAN OIL HISTORY :

- ❖ Oil Seepages observed by British Army officers on survey expeditions in Brahmaputra Valley -Lt. R. Wilcox of the 46th Regiment Native Infantry (1826), Major A. White and Captain P.S. Hannay.
- ❖ H.B Medlicott 1865 - In Upper Assam started Assam Railways & Trading Company
- ❖ Oil drilling undertaken in Makum-Namdang area. 1866.. Digboi field discovered when Oil was struck at a depth of 172 feet in 1889
- ❖ Digboi refinery started in 1902
- ❖ 1935 - Naharkotiya- the first oil field discovered by Geo-physical methods
- ❖ BOC- Burmah Oil Company reign supreme with CALTEX and STANDARD VACUUM.
- ❖ 1954 - Keshava Dev Malaviya Junior Minister in Nehru's cabinet.
- ❖ 1955 - petroleum division in GSI
- ❖ 14 Aug 1956- Oil & Natural Gas Commission
- ❖ Malaviya, Minister of Natural resources and Scientific Research - first Chairman.
- ❖ An eminent Russian geologist, Prof N.A. Kalinin, accompanied by drilling consultants landed in Delhi in Dec. 1955.
- ❖ 1957 Jwalamukhi drilling started with a Romanian oil rig.
- ❖ ONGC strikes Black Gold-1958 at Cambay
- ❖ Discovers oil at Rudrasagar, Assam, in 1960
- ❖ Discovers commercial oil at Ankleswar in 1960.
- ❖ 1961: ONGC acquires SS Mahindra, the seismic survey vessel
- ❖ ONGC goes International in 1965: Hydrocarbons India Ltd starts its operations in Iran
- ❖ 1970: operation leapfrog , at Aliabet- First offshore venture
- ❖ 1973: ONGC acquires Sagar Samrat.
- ❖ 1974: Bombay High discovered
- ❖ 1976: production from Bombay high commences on 21st May 1976
- ❖ 1985: production touches a peak of 35 mmt.
- ❖ 1989: South discoveries after four forays.
- ❖ 1994: ONGC becomes a Corporation
- ❖ 2001: Redevelopment plan of Bombay High launched.
- ❖ 2003: Sagar Sammriddhi- biggest deepwater campaign by any single operator, launched.
- ❖ 2003 ONGC acquires MRPL - an integrated oil company
- ❖ ONGC TODAY: Largest corporate in India.; Third Largest E&P company in the world.; Greenest company of India.; Greatest wealth creator in India.; New Businesses: Clean coal technologies, Gas hydrates, Wind power, Refineries, Retailing....Drilling to Dispensing.